

AI STUDYBUDDY API

Scalability & Future Plan

Recommended reliability, security, and learning improvements

TEAM ID	SWTID-2026-2776
TEAM SIZE	5
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Near-term fixes

- Only assign admin role through trusted provisioning; ignore/reject role in public registration.
- Enforce owner checks on summarize, flashcards, quiz, and study-plan routes.
- Extract text from PDFs using a real PDF parser; validate MIME and file signatures.
- Use try/finally cleanup for temporary uploads.
- Add tests and validate generated JSON with schemas.

Growth

Add pagination, indexes, background jobs for long AI operations, AI rate limits, request limits, timeouts, monitoring, and managed secret storage.

Learning improvements

Allow regeneration controls, progress tracking, topic tagging, export, and user feedback after validating student needs.

Privacy

Disclose that material text is sent to Gemini; minimize sensitive content and define retention/deletion rules.

Boundary

These are recommendations; these capabilities are not shown in the submitted code.